ENGR 4333/5333: Digital Signal Processing

Continuous-Time Analog Filters

Chapter 1 and 2

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Outline Review

- Fourier Series (Section 1.7)
- Fourier Transform (Section 1.8)
- Laplace Transform (Section 1.10)
- Frequency Response of and LTIC System (Section 2.1)
- Signal Transmission through LTIC Systems (Section 2.2)

1.7
The Fourier Series

The Fourier Series

The exponential Fourier series representation of a periodic signal with fundamental frequency ω_o.

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t}$$

Note:
X_k is equivalent to D_n in signals and systems.

The Fourier series coefficient X_k.

$$X_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_o t} dt$$

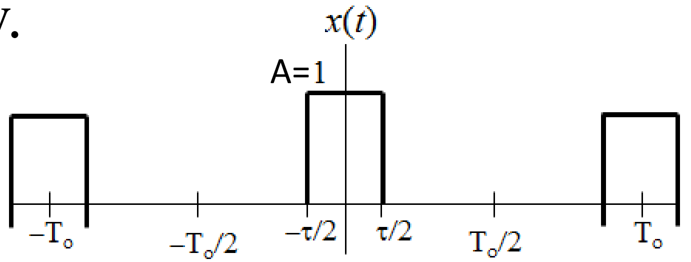
The compact trigonometric Fourier series representation of periodic.

$$x(t) = C_0 + \sum_{k=1}^{\infty} C_k \cos(k\omega_o t + \theta_k)$$

where $C_0 = X_0$, $C_k = 2|X_k|$, and $\theta_k = \angle X_k$

Fourier Series of Pulse Train

Find the Fourier series representation of the periodic signal $x(t)$ below.

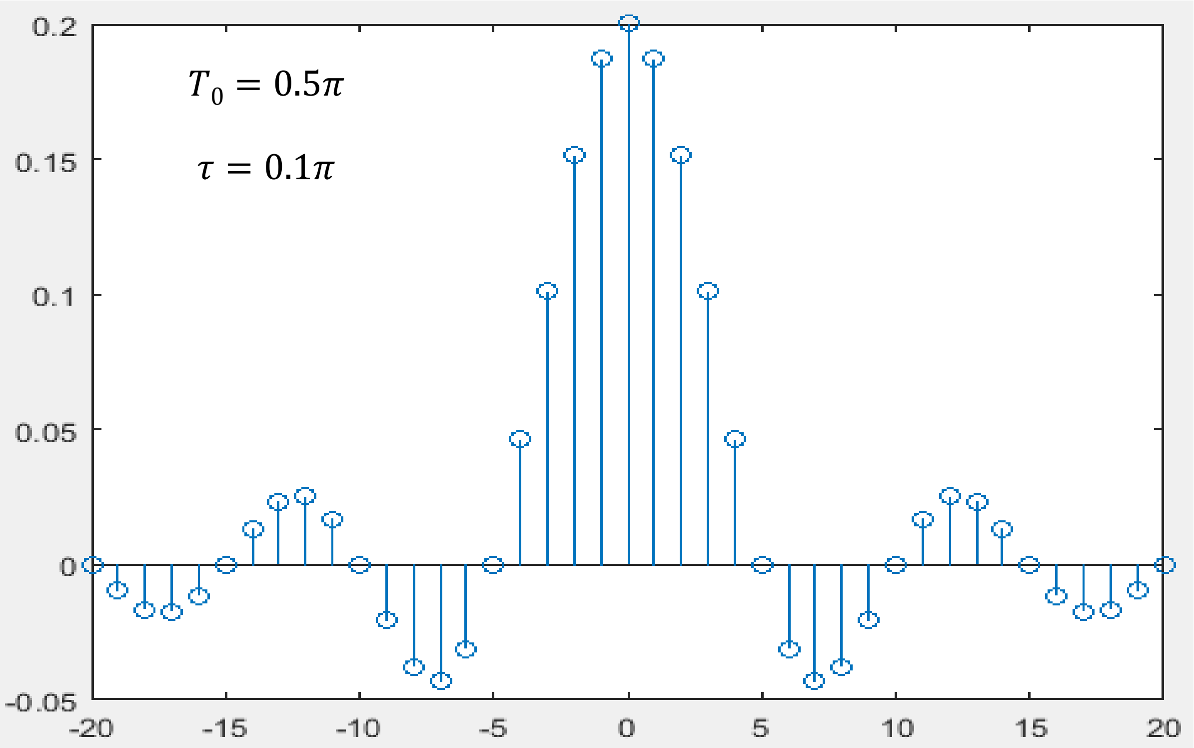


Answer:

$$X_k = \frac{A\tau}{T_0} sinc\left(\frac{k\omega_o\tau}{2\pi}\right)$$

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{A\tau}{T_0} sinc\left(\frac{k\omega_o\tau}{2\pi}\right) e^{jk\omega_o t}$$

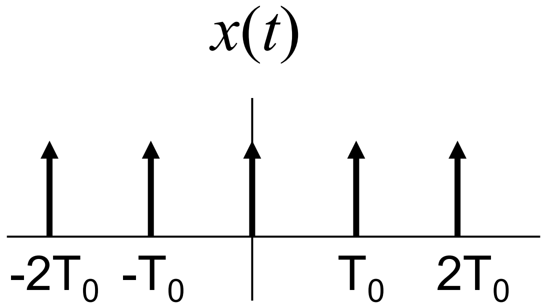
$$x(t) = \frac{A\tau}{T_0} + \sum_{k=1}^{\infty} \frac{2A\tau}{T_0} sinc\left(\frac{k\omega_o\tau}{2\pi}\right) \cos(k\omega_o t + \theta_k$$



$$sinc(x) = \frac{\sin(\pi x)}{\pi x}$$

Fourier Series of Impulse Train

Find the Fourier series representation of the periodic signal $x(t) = \delta_{T_0}(t)$.



Answer:

$$X_k = \frac{1}{T_0}$$

$$x(t) = \frac{1}{T_0} \sum_{k=-\infty}^{\infty} e^{jk\omega_o t}$$

$$x(t) = \frac{1}{T_0} + \sum_{k=1}^{\infty} \frac{2}{T_0} \cos(k\omega_o t)$$

1.8
The Fourier Transform

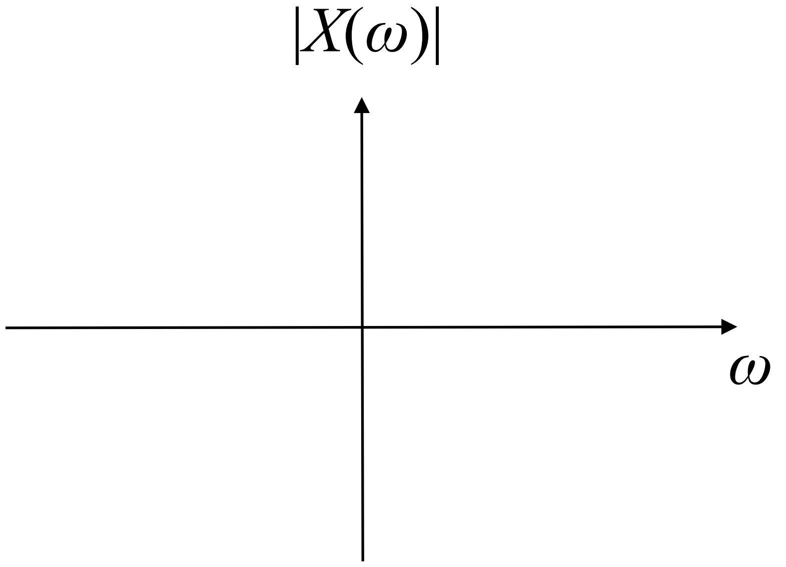
Fourier Transform

The Fourier transform of a signal represent the spectrum of the signal.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

The inverse Fourier transform of the spectrum of the signal.

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

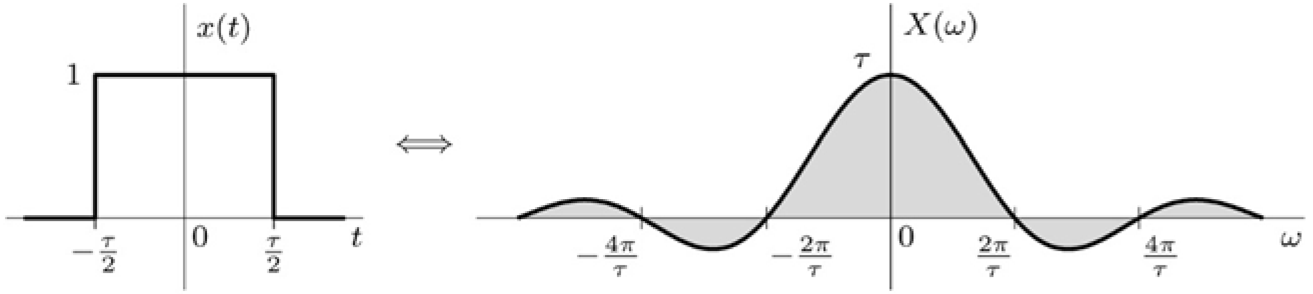


Fourier Transform of a Rectangular Pulse

Find the Fourier transform of the rectangular pulse $x(t) = \Pi\left(\frac{t}{\tau}\right)$

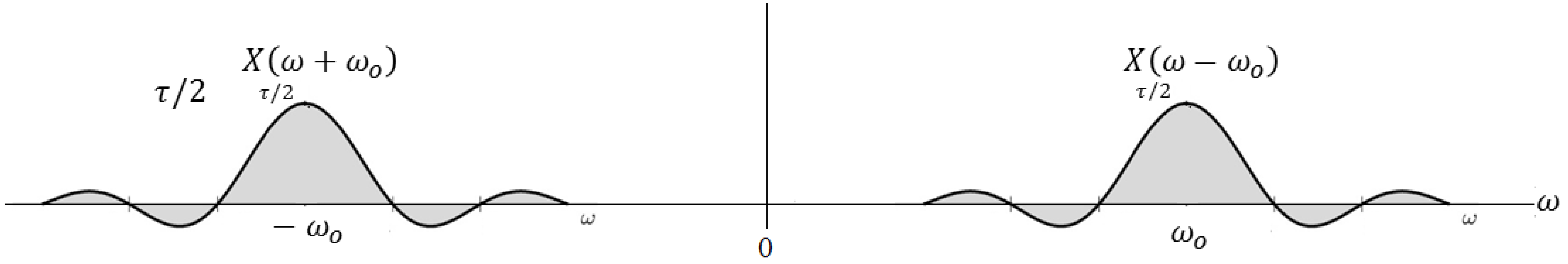
Answer:

$$X(\omega) = \tau sinc\left(\frac{\tau\omega}{2\pi}\right)$$



Modulation Property: $x(t)\cos(\omega_o t) \xleftrightarrow{FT} \frac{1}{2}[X(\omega+\omega_o) + X(\omega-\omega_o)]$

$$\Pi\left(\frac{t}{\tau}\right)\cos(\omega_o t)$$



Sampling Theorem: $x_s(t) = x(t)\delta_{T_0}(t) = \frac{1}{T_0}x(t) + \sum_{k=1}^{\infty}\frac{2}{T_0}\cos(k\omega_o t)x(t)$

What is the spectrum of a sampled rectangular pulse?

	$x(t)$	$X(\omega)$	
1.	$e^{\lambda t}u(t)$	$\frac{1}{j\omega-\lambda}$	$\mathrm{Re}\{\lambda\}<0$
2.	$e^{\lambda t}u(-t)$	$-\frac{1}{j\omega-\lambda}$	$\mathrm{Re}\{\lambda\}>0$
3.	$e^{\lambda\vert t\vert}$	$\frac{-2\lambda}{\omega^2+\lambda^2}$	$\mathrm{Re}\{\lambda\}<0$
4.	$t^k e^{\lambda t}u(t)$	$\frac{k!}{(j\omega-\lambda)^{k+1}}$	$\mathrm{Re}\{\lambda\}<0$
5.	$e^{-at}\cos(\omega_0 t)u(t)$	$\frac{a+j\omega}{(a+j\omega)^2+\omega_0^2}$	$a>0$
6.	$e^{-at}\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(a+j\omega)^2+\omega_0^2}$	$a>0$
7.	$\Pi\left(\frac{t}{\tau}\right)$	$\tau\mathrm{sinc}\left(\frac{\tau\omega}{2\pi}\right)$	$\tau>0$
8.	$\frac{B}{\pi}\mathrm{sinc}\left(\frac{Bt}{\pi}\right)$	$\Pi\left(\frac{\omega}{2B}\right)$	$B>0$
9.	$\Lambda\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2}\mathrm{sinc}^2\left(\frac{\tau\omega}{4\pi}\right)$	$\tau>0$
10.	$\frac{B}{2\pi}\mathrm{sinc}^2\left(\frac{Bt}{2\pi}\right)$	$\Lambda\left(\frac{\omega}{2B}\right)$	$B>0$
11.	$e^{-t^2/2\sigma^2}$	$\sigma\sqrt{2\pi}e^{-\sigma^2\omega^2/2}$	$\sigma>0$
12.	$\delta(t)$	1	
13.	1	$2\pi\delta(\omega)$	
14.	$u(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$	
15.	$\mathrm{sgn}(t)$	$\frac{2}{j\omega}$	
16.	$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$	
17.	$\cos(\omega_0 t)$	$\pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]$	
18.	$\sin(\omega_0 t)$	$\frac{\pi}{j}\left[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\right]$	
19.	$\cos(\omega_0 t)u(t)$	$\frac{\pi}{2}\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]+\frac{j\omega}{\omega_0^2-\omega^2}$	
20.	$\sin(\omega_0 t)u(t)$	$\frac{\pi}{2j}\left[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\right]+\frac{\omega_0}{\omega_0^2-\omega^2}$	
21.	$\sum_{k=-\infty}^{\infty}\delta(t-kT)$	$\omega_0\sum_{k=-\infty}^{\infty}\delta(\omega-k\omega_0)$	$\omega_0=\frac{2\pi}{T}$

Synthesis: $x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}X(\omega)e^{j\omega t}\,d\omega$	**Synthesis:** $x(t)=\sum_{-\infty}^{\infty}X_k e^{jk\omega_0 t}$
Analysis: $X(\omega)=\int_{-\infty}^{\infty}x(t)e^{-j\omega t}\,dt$	**Analysis:** $X_k=\frac{1}{T_0}\int_{T_0}x(t)e^{-jk\omega_0 t}\,dt$
Duality: if $x(t)\Longleftrightarrow X(\omega)$, then $X(t)\Longleftrightarrow 2\pi x(-\omega)$	**Duality:**
Linearity: $ax(t)+by(t)\Longleftrightarrow aX(\omega)+bY(\omega)$	**Linearity:** $ax(t)+by(t)\Longleftrightarrow aX_k+bY_k$
Complex Conjugation: $x^*(t)\Longleftrightarrow X^*(-\omega)$	**Complex Conjugation:** $x^*(t)\Longleftrightarrow X^*_{-k}$
Scaling and Reversal: $x(at)\Longleftrightarrow\frac{1}{\vert a\vert}X\left(\frac{\omega}{a}\right)$ $x(-t)\Longleftrightarrow X(-\omega)$	**Scaling and Reversal:** $x(-t)\Longleftrightarrow X_{-k}$
Shifting: $x(t-t_0)\Longleftrightarrow X(\omega)e^{-j\omega t_0}$ $x(t)e^{j\omega_0 t}\Longleftrightarrow X(\omega-\omega_0)$	**Shifting:** $x(t-t_0)\Longleftrightarrow X_k e^{-jk\omega_0 t_0}$ $x(t)e^{jk_0\omega_0 t}\Longleftrightarrow X_{k-k_0}$
Differentiation: $\frac{d}{dt}x(t)\Longleftrightarrow j\omega X(\omega)$ $-jtx(t)\Longleftrightarrow\frac{d}{d\omega}X(\omega)$	**Differentiation:** $\frac{d}{dt}x(t)\Longleftrightarrow jk\omega_0 X_k$
Time Integration: $\int_{-\infty}^{t}x(\tau)d\tau\Longleftrightarrow\frac{X(\omega)}{j\omega}+\pi X(0)\delta(\omega)$	**Time Integration:**
Convolution: $x(t)*y(t)\Longleftrightarrow X(\omega)Y(\omega)$ $x(t)y(t)\Longleftrightarrow\frac{1}{2\pi}X(\omega)*Y(\omega)$	**Convolution:** $\frac{1}{T_0}x(t)\circledast y(t)\Longleftrightarrow X_kY_k$ $x(t)y(t)\Longleftrightarrow X_k*Y_k$
Correlation: $\rho_{x,y}(\tau)=x(\tau)*y^*(-\tau)\Longleftrightarrow X(\omega)Y^*(\omega)$	**Correlation:** $\rho_{x,y}(\tau)=\frac{1}{T_0}x(\tau)\circledast y^*(-\tau)\Longleftrightarrow X_kY_k^*$
Parseval's: $E_x=\int_{-\infty}^{\infty}\vert x(t)\vert^2\,dt=\frac{1}{2\pi}\int_{-\infty}^{\infty}\vert X(\omega)\vert^2\,d\omega$	**Parseval's:** $P_x=\frac{1}{T_0}\int_{T_0}\vert x(t)\vert^2\,dt=\sum_{k=-\infty}^{\infty}\vert X_k\vert^2$

1.10
The Laplace Transform

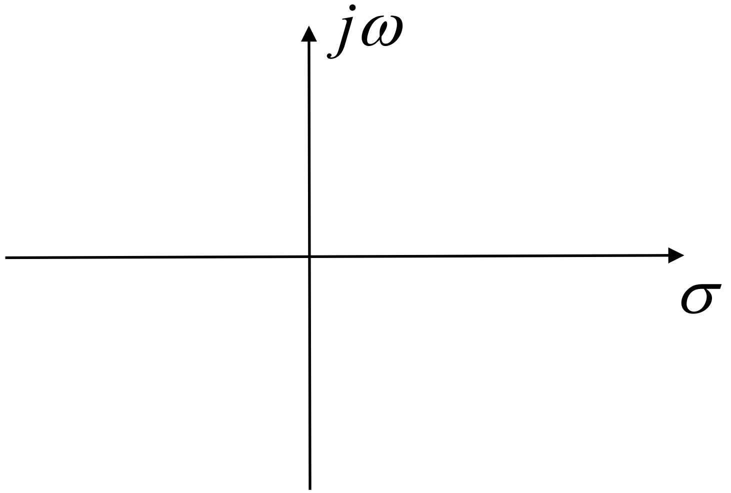
The Laplace Transform

The Laplace transform of a signal represent the complex spectrum of the signal.

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

Where s is the complex frequency $s = \sigma + j\omega$.

s-plane: $|X(s)|$



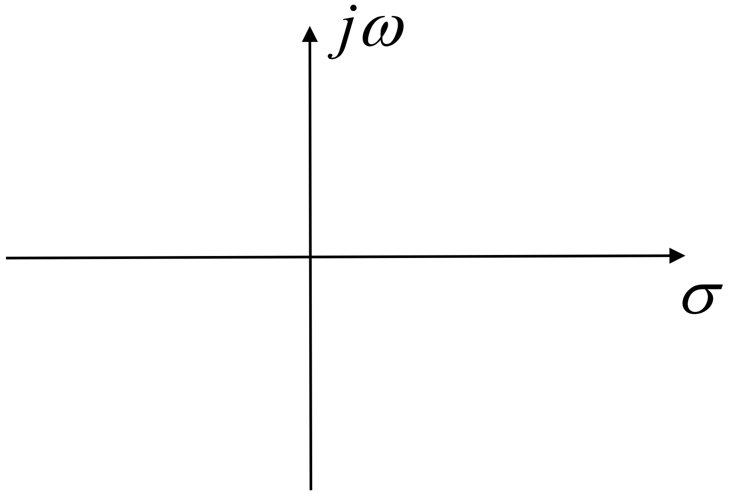
The inverse Laplace transform of $X(s)$.

$$x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}ds$$

Use partial fraction expansion and the table to find the inverse Laplace transform

The Laplace Transform of a Unit Step Function

s-plane: $|X(s)|$



Find the Laplace transform of $x(t) = u(t)$

Answer:

$$X(s) = \frac{1}{s} \qquad \text{ROC Re}\{s\} > 0$$

Note: $X(\omega)$ can be obtained from $X(s)$ by setting $\sigma = 0$ and the ROC of $X(s)$ include the jω axis.

$$e^{\lambda t}u(t) \overset{LT}{\longleftrightarrow} \frac{1}{s-\lambda}$$

$$\text{Re}\{s\} > \lambda$$

$$e^{\lambda t}u(t) \overset{FT}{\longleftrightarrow} \frac{1}{j\omega-\lambda}$$

	$x(t)$	$X(s)$	ROC
1.	$\delta(t)$	1	All s
2.	$u(t)$	$\frac{1}{s}$	$\mathrm{Re}\{s\} > 0$
3.	$t^k u(t)$	$\frac{k!}{s^{k+1}}$	$\mathrm{Re}\{s\} > 0$
4.	$e^{\lambda t}u(t)$	$\frac{1}{s-\lambda}$	$\mathrm{Re}\{s\} > \mathrm{Re}\{\lambda\}$
5.	$t^k e^{\lambda t}u(t)$	$\frac{k!}{(s-\lambda)^{k+1}}$	$\mathrm{Re}\{s\} > \mathrm{Re}\{\lambda\}$
6.	$\cos(\omega_0 t)u(t)$	$\frac{s}{s^2+\omega_0^2}$	$\mathrm{Re}\{s\} > 0$
7.	$\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\mathrm{Re}\{s\} > 0$
8.	$e^{-at}\cos(\omega_0 t)u(t)$	$\frac{s+a}{(s+a)^2+\omega_0^2}$	$\mathrm{Re}\{s\} > -a$
9.	$e^{-at}\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(s+a)^2+\omega_0^2}$	$\mathrm{Re}\{s\} > -a$
10.	$e^{-at}\cos(\omega_0 t+\theta)\,u(t)$	$\frac{\cos(\theta)s+a\cos(\theta)-\omega_0\sin(\theta)}{s^2+2as+(a^2+\omega_0^2)}$ $= \frac{0.5e^{j\theta}}{s+a-j\omega_0} + \frac{0.5e^{-j\theta}}{s+a+j\omega_0}$	$\mathrm{Re}\{s\} > -a$
11.	$u(-t)$	$-\frac{1}{s}$	$\mathrm{Re}\{s\} < 0$
12.	$t^k u(-t)$	$-\frac{k!}{s^{k+1}}$	$\mathrm{Re}\{s\} < 0$
13.	$e^{\lambda t}u(-t)$	$-\frac{1}{s-\lambda}$	$\mathrm{Re}\{s\} < \mathrm{Re}\{\lambda\}$
14.	$t^k e^{\lambda t}u(-t)$	$-\frac{k!}{(s-\lambda)^{k+1}}$	$\mathrm{Re}\{s\} < \mathrm{Re}\{\lambda\}$
15.	$\cos(\omega_0 t)u(-t)$	$-\frac{s}{s^2+\omega_0^2}$	$\mathrm{Re}\{s\} < 0$
16.	$\sin(\omega_0 t)u(-t)$	$-\frac{\omega_0}{s^2+\omega_0^2}$	$\mathrm{Re}\{s\} < 0$
17.	$e^{-at}\cos(\omega_0 t)u(-t)$	$-\frac{s+a}{(s+a)^2+\omega_0^2}$	$\mathrm{Re}\{s\} < -a$
18.	$e^{-at}\sin(\omega_0 t)u(-t)$	$-\frac{\omega_0}{(s+a)^2+\omega_0^2}$	$\mathrm{Re}\{s\} < -a$
19.	$e^{-at}\cos(\omega_0 t+\theta)\,u(-t)$	$-\frac{\cos(\theta)s+a\cos(\theta)-\omega_0\sin(\theta)}{s^2+2as+(a^2+\omega_0^2)}$ $= -\frac{0.5e^{j\theta}}{s+a-j\omega_0} + \frac{0.5e^{-j\theta}}{s+a+j\omega_0}$	$\mathrm{Re}\{s\} < -a$

Bilateral Laplace Transform	Unilateral Laplace Transform
Synthesis: $x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}\,ds$	**Synthesis:** $x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}\,ds$
Analysis: $X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt$, ROC: R_x	**Analysis:** $X(s) = \int_{0^-}^{\infty} x(t)e^{-st}\,dt$
Linearity: $ax(t)+by(t) \overset{\mathcal{L}}{\Longleftrightarrow} aX(s)+bY(s)$, ROC: At least $R_x \cap R_y$	**Linearity:** $ax(t)+by(t) \overset{\mathcal{L}_u}{\Longleftrightarrow} aX(s)+bY(s)$
Complex Conjugation: $x^*(t) \overset{\mathcal{L}}{\Longleftrightarrow} X^*(s^*)$, ROC: R_x	**Complex Conjugation:** $x^*(t) \overset{\mathcal{L}_u}{\Longleftrightarrow} X^*(s^*)$
Scaling and Reversal: $x(at) \overset{\mathcal{L}}{\Longleftrightarrow} \frac{1}{\lvert a\rvert}X\left(\frac{s}{a}\right)$, ROC: R_x scaled by $1/a$ $x(-t) \overset{\mathcal{L}}{\Longleftrightarrow} X(-s)$, ROC: R_x reflected	**Scaling and Reversal:** If $a>0$: $x(at) \overset{\mathcal{L}_u}{\Longleftrightarrow} \frac{1}{a}X\left(\frac{s}{a}\right)$
Shifting: $x(t-t_0) \overset{\mathcal{L}}{\Longleftrightarrow} X(s)e^{-st_0}$, ROC: R_x $x(t)e^{s_0 t} \overset{\mathcal{L}}{\Longleftrightarrow} X(s-s_0)$, ROC: R_x shifted by $\mathrm{Re}\{s_0\}$	**Shifting:** If $t_0>0$: $x(t-t_0) \overset{\mathcal{L}_u}{\Longleftrightarrow} X(s)e^{-st_0}$ $x(t)e^{s_0 t} \overset{\mathcal{L}_u}{\Longleftrightarrow} X(s-s_0)$
Differentiation: $\frac{d}{dt}x(t) \overset{\mathcal{L}}{\Longleftrightarrow} sX(s)$, ROC: At least R_x $-tx(t) \overset{\mathcal{L}}{\Longleftrightarrow} \frac{d}{ds}X(s)$, ROC: R_x	**Differentiation:** $\frac{d}{dt}x(t) \overset{\mathcal{L}_u}{\Longleftrightarrow} sX(s)-x(0^-)$ (general case shown below) $-tx(t) \overset{\mathcal{L}_u}{\Longleftrightarrow} \frac{d}{ds}X(s)$
Time Integration: $\int_{-\infty}^{t} x(\tau)d\tau \overset{\mathcal{L}}{\Longleftrightarrow} \frac{1}{s}X(s)$, ROC: At least $R_x \cap (\mathrm{Re}\{s\}>0)$	**Time Integration:** $\int_{0^-}^{t} x(\tau)d\tau \overset{\mathcal{L}_u}{\Longleftrightarrow} \frac{1}{s}X(s)$
Convolution: $x(t)*y(t) \overset{\mathcal{L}}{\Longleftrightarrow} X(s)Y(s)$, ROC: At least $R_x \cap R_y$	**Convolution:** $x(t)*y(t) \overset{\mathcal{L}_u}{\Longleftrightarrow} X(s)Y(s)$

Unilateral Laplace Transform Time Differentiation, General Case

$$x^{(k)}(t) = \frac{d^k}{dt^k}x(t) \overset{\mathcal{L}_u}{\Longleftrightarrow} s^k X(s) - \sum_{i=0}^{k-1} s^{k-1-i}x^{(i)}(0^-)$$

2.1
Frequency Response of an LTIC System

Frequency Response of an LTIC System

Time Domain

$$\delta(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)=h(t)$$

$$x(t) = e^{j\omega_o t} \qquad\qquad y(t) = x(t) * h(t)$$

Laplace Domain

$$X(s) \longrightarrow \boxed{H(s)} \longrightarrow Y(s) = X(s)\,H(s)$$

$$x(t) = e^{s_o t} \qquad\qquad y(t) = H(s_o)e^{s_o t}$$

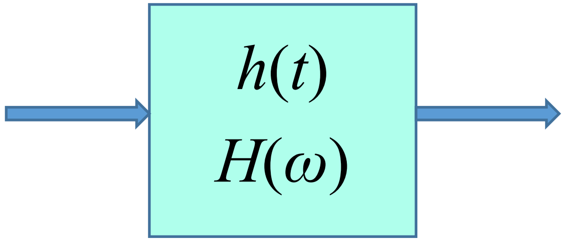
Frequency Domain

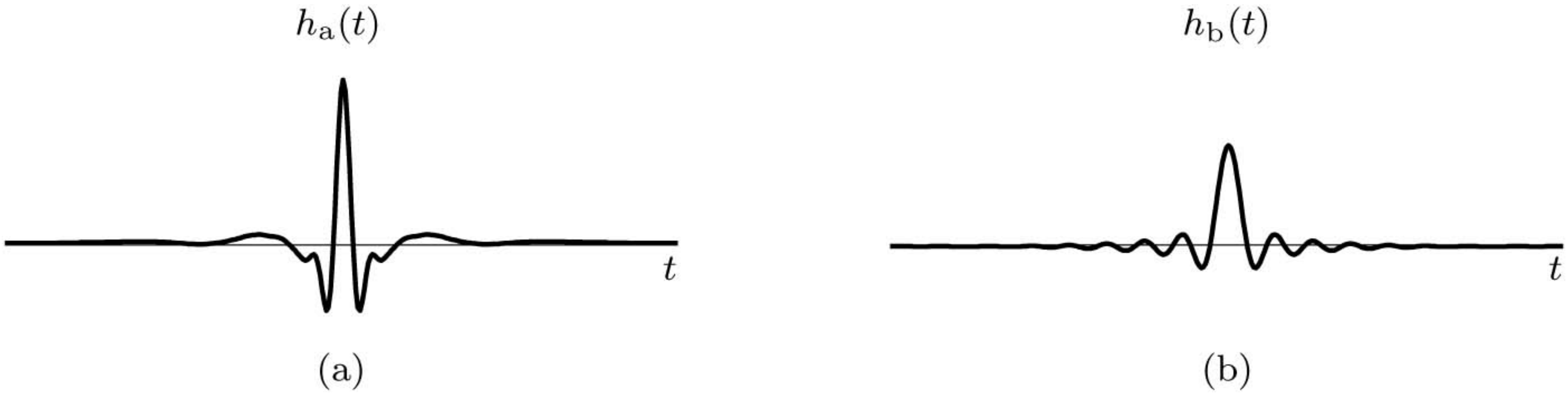
$$X(\omega) \longrightarrow \boxed{H(\omega)} \longrightarrow Y(\omega) = X(\omega)\,H(\omega)$$

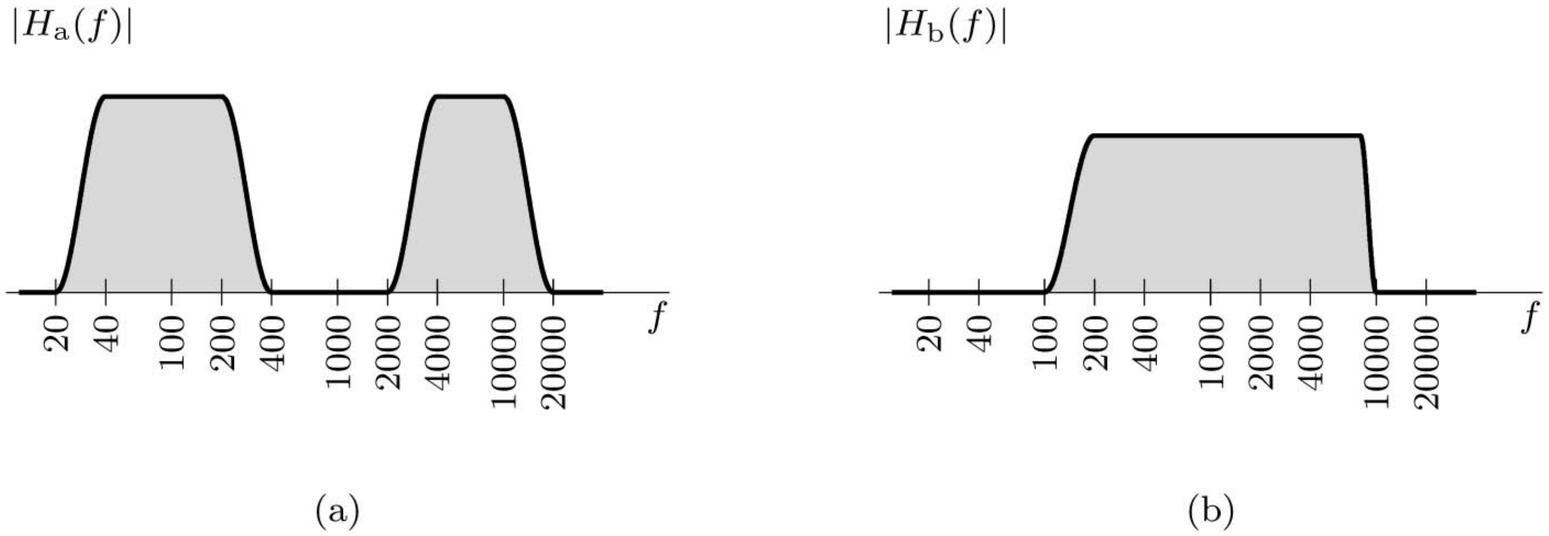
$$x(t) = e^{j\omega_o t} \qquad\qquad y(t) = H(\omega_o)e^{j\omega_o t}$$

$$x(t) = A\cos(\omega_o t + \theta) \qquad\qquad y(t) = [H(\omega_o)]A\cos\left(\omega_o t + \theta + \angle H(\omega_o)\right)$$

Time Domain Verses Frequency Domain







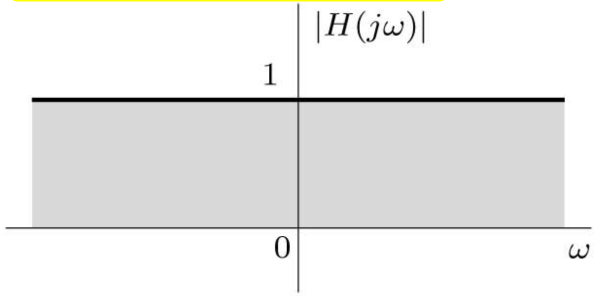
Frequency Response of Common Systems

An Ideal Time Delay of T seconds

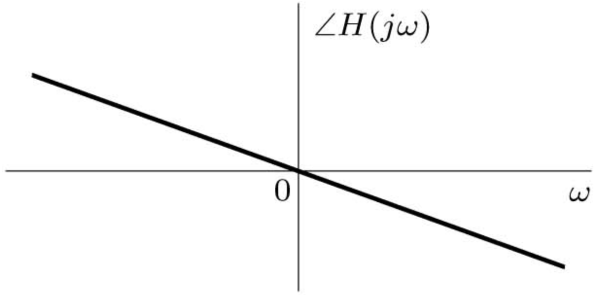
$$y(t) = x(t - T)$$

$$H(\omega) = e^{-jT\omega}$$

$$|H(j\omega)| = 1 \quad \text{and} \quad \angle H(j\omega) = -\omega T$$



(a)



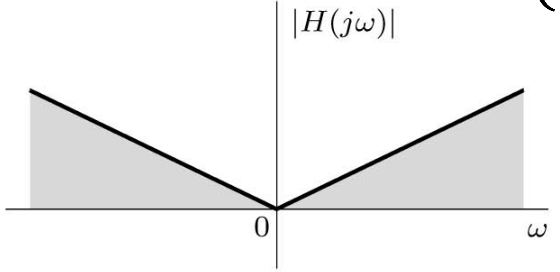
(b)

An Ideal Differentiator

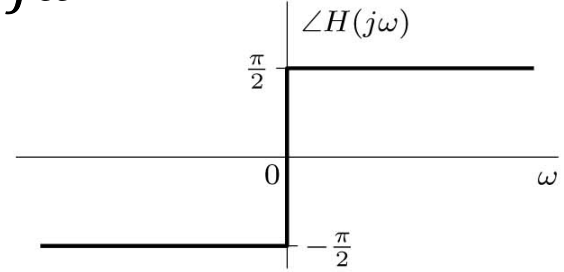
$$y(t) = \frac{dx}{dt}$$

$$H(\omega) = j\omega$$

$$|H(j\omega)| = |\omega| \quad \text{and} \quad \angle H(j\omega) = \frac{\pi}{2}\text{sgn}(\omega)$$



(a)



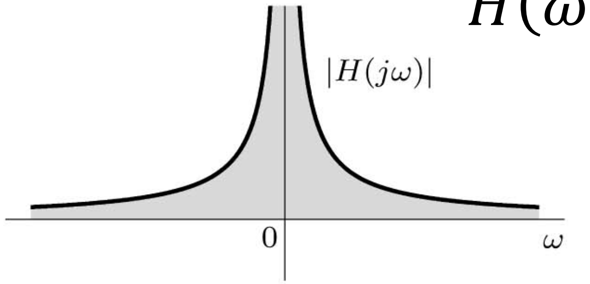
(b)

An Ideal Integrator

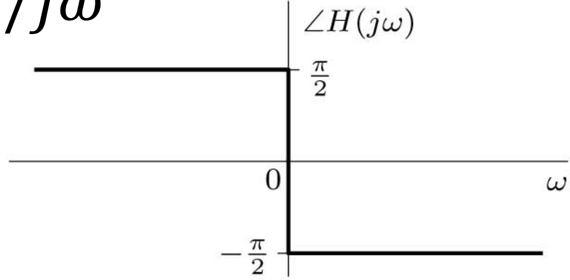
$$y(t) = \int x(\tau)d\tau$$

$$H(\omega) = 1/j\omega$$

$$|H(j\omega)| = \frac{1}{|\omega|} \quad \text{and} \quad \angle H(j\omega) = -\frac{\pi}{2}\text{sgn}(\omega)$$



(a)



(b)

Pole-Zero Plots

LTIC system is described by the constant-coefficient linear differential Equation.

$$a_K \frac{d^K y}{dt^K} + a_{K-1} \frac{d^{K-1} y}{dt^{K-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_L \frac{d^L x}{dt^L} + b_{L-1} \frac{d^{L-1} x}{dt^{L-1}} + \cdots + b_1 \frac{dx}{dt} + b_0 x$$

$$H(s) = \frac{B(s)}{A(s)} = \frac{b_L s^L + b_{L-1} s^{L-1} + \cdots + b_1 s + b_0}{a_K s^K + a_{K-1} s^{K-1} + \cdots + a_1 s + a_0} = \left(\frac{b_L}{a_K}\right) \frac{\prod_{l=1}^{L} (s - z_l)}{\prod_{k=1}^{K} (s - p_k)}$$

Example:

Find the transfer function $H(s)$ and the frequency response $H(\omega)$ of the system described by the differential equation

$$\frac{d^2 y}{dt^2} + 2\frac{dy}{dt} + 5y = 2\frac{dx}{xt}$$

Answer: $H(s) = \dfrac{2s}{s^2 + 2s + 5}$

Example

Consider a system whose transfer function is $H(s) = \dfrac{2s}{s^2 + 2s + 5}$

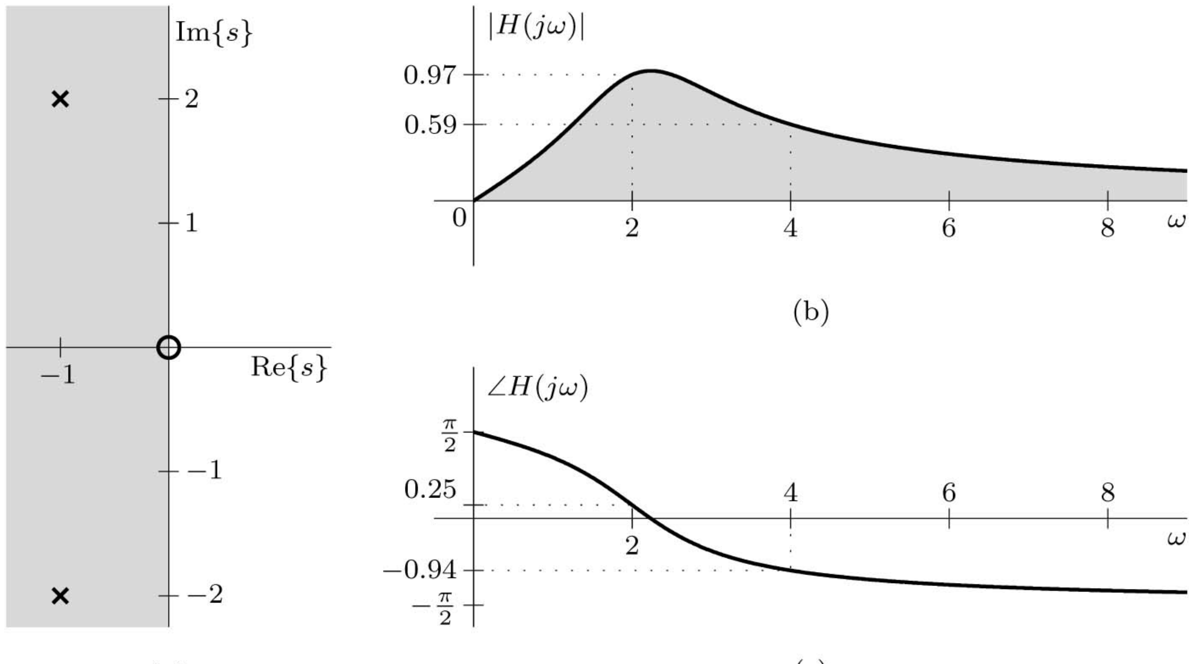
Determine and plot the poles and zeros of $H(s)$, and use the pole-zero information to predict overall system behavior. Confirm this predicted behavior by computing and graphing the system's frequency response (magnitude and phase). Lastly, find and plot the system response $y(t)$ for the inputs **(a)** $x_a(t) = \cos(2t)$ **(b)** $x_b(t) = 2\sin(4t - \pi/3)$

$$|H(j\omega)| = \frac{\sqrt{(2\omega)^2}}{\sqrt{(5-\omega^2)^2 + (2\omega)^2}}$$

$$\angle H(j\omega) = \mathrm{sgn}(\omega)\frac{\pi}{2} - \tan^{-1}\left(\frac{2\omega}{5-\omega^2}\right)$$

$$y_a(t) = 0.9701\cos(2t + 0.2450)$$

$$\begin{aligned} y_b(t) &= 2(0.5882)\sin(4t - \pi/3 - 0.9420) \\ &= 1.1763\sin(4t - 1.9892) \end{aligned}$$



(a) (b) (c)

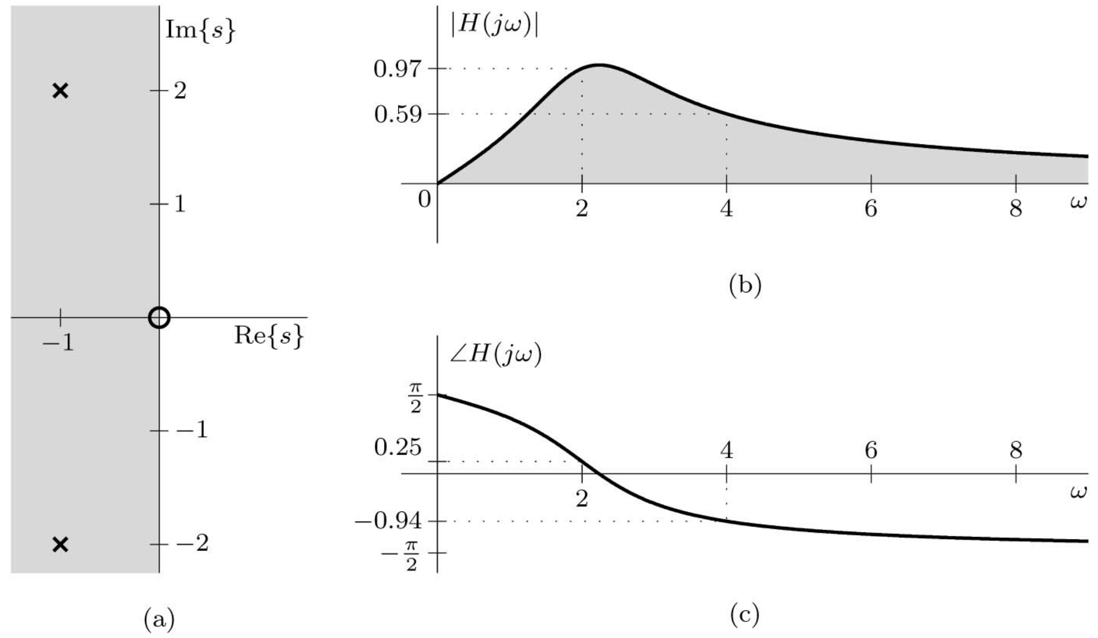
Continue Example

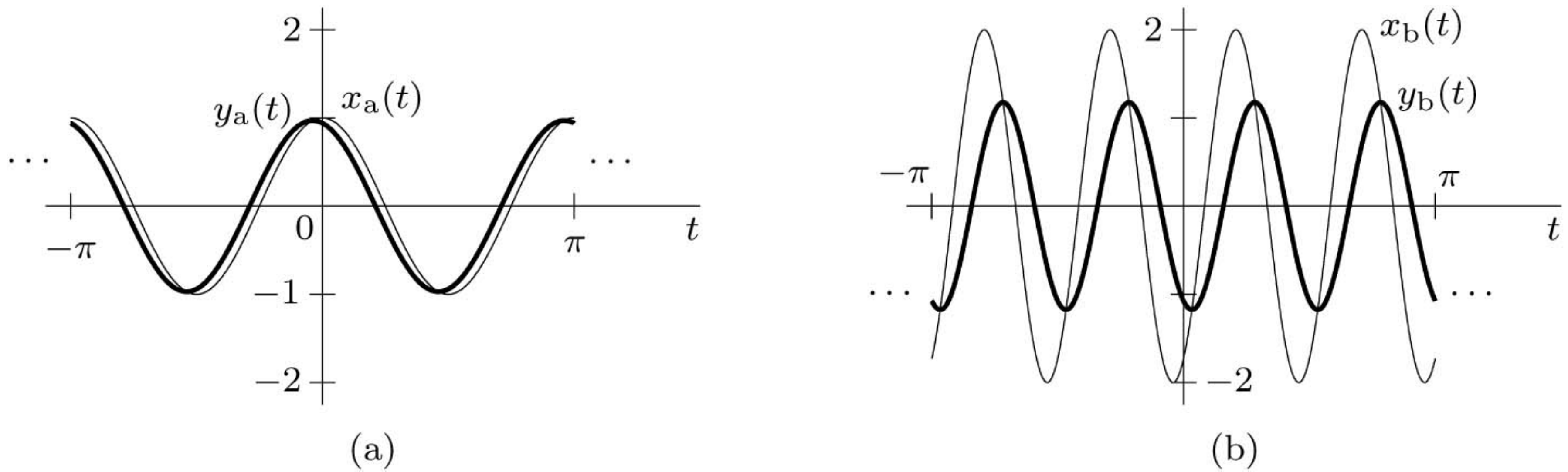
For input $x_a(t) = \cos(2t)$

$$y_a(t) = 0.9701\cos(2t + 0.2450)$$

For input $x_b(t) = 2\sin(4t - \pi/3)$

$$y_b(t) = 2(0.5882)\sin(4t - \pi/3 - 0.9420)$$
$$= 1.1763\sin(4t - 1.9892)$$





2.4
Data Truncation by Windows

Data Truncation by Windows

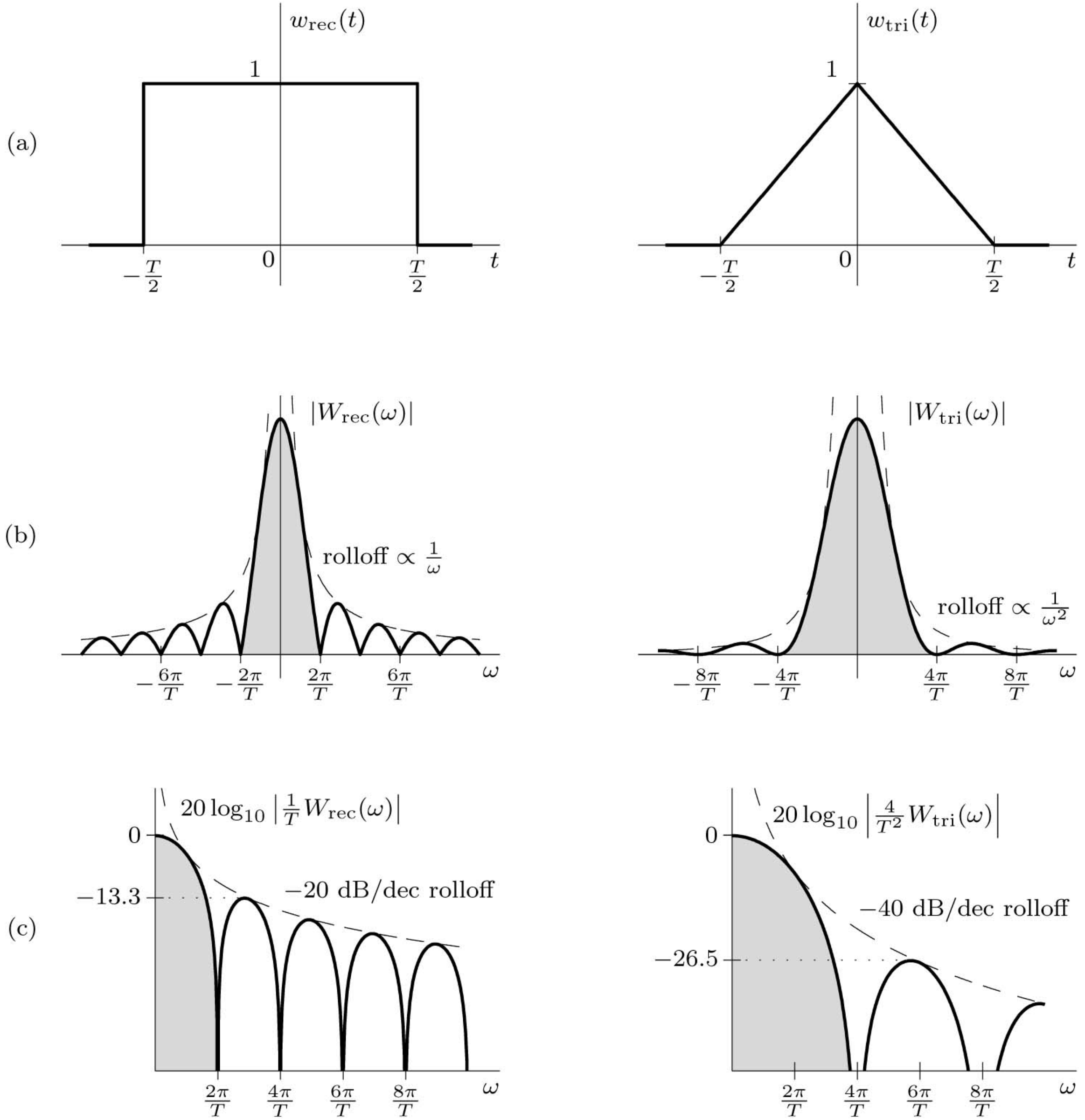
$x(t)$: is the signal
$w(t)$: is the window
$x_w(t)$: is the truncated $x(t)$.

The two side effects of truncation is spectral spreading and spectral leakage.

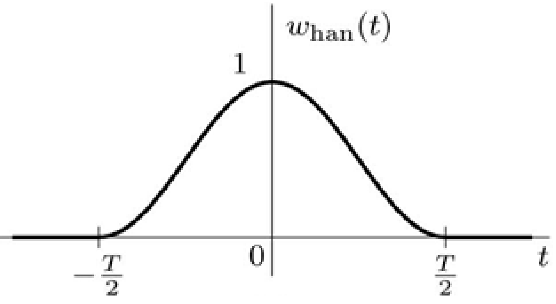
$$x_w(t) = x(t)w(t)$$

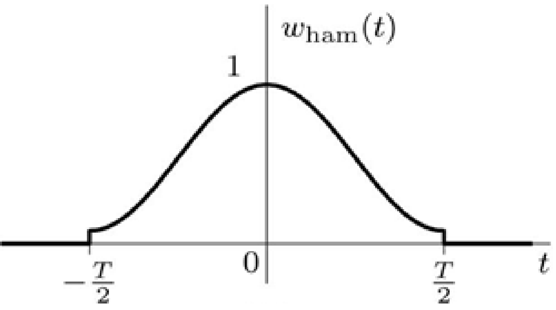
$$X_w(\omega) = \frac{1}{2\pi}X(\omega) * W(\omega)$$

Example: What is the effect of truncating a dc signal?



Common Window Functions





	Window $w(t)$	Main Lobe Width	Rolloff Rate [dB/dec]	Peak Side Lobe Level [dB]
1.	Rectangular: $\Pi\left(\frac{t}{T}\right)$	$\frac{4\pi}{T}$	-20	-13.3
2.	Triangular (Bartlett): $\Lambda\left(\frac{t}{T}\right)$	$\frac{8\pi}{T}$	-40	-26.5
3.	Hann: $\frac{1}{2}\left[1+\cos\left(\frac{2\pi t}{T}\right)\right]\Pi\left(\frac{t}{T}\right)$	$\frac{8\pi}{T}$	-60	-31.5
4.	Hamming: $\left[0.54+0.46\cos\left(\frac{2\pi t}{T}\right)\right]\Pi\left(\frac{t}{T}\right)$	$\frac{8\pi}{T}$	-20	-42.7
5.	Blackman: $\left[0.42+0.5\cos\left(\frac{2\pi t}{T}\right)+0.08\cos\left(\frac{4\pi t}{T}\right)\right]\Pi\left(\frac{t}{T}\right)$	$\frac{12\pi}{T}$	-60	-58.1
6.	Kaiser: $\frac{I_0\left(\alpha\sqrt{1-4\left(\frac{t}{T}\right)^2}\right)}{I_0(\alpha)}\Pi\left(\frac{t}{T}\right)$	varies with α	-20	varies with α